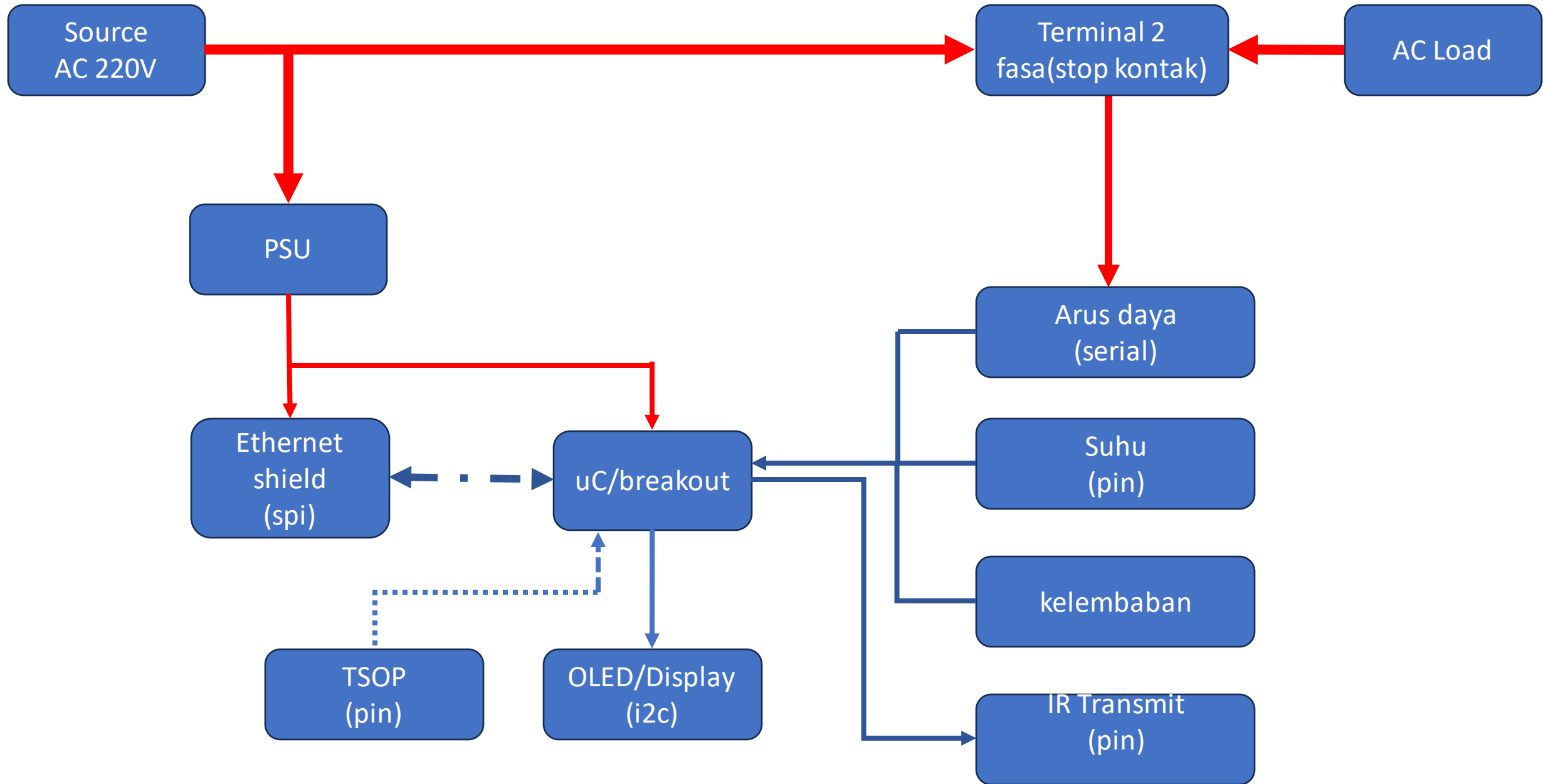


PT. Sarana Makin Mulia
Departemen Engineering

Projek Monitoring Room

Wattmeter, temperature, humidity & control AC



Step

1. Pof controlling AC with Arduino (IR,tsop,oled)hack
 1. Read data universal remote
 2. Send data from ir led to ac
2. Read current AC from sensor
3. Pof connect to internet(database)
 1. Create simple database & send data
 2. Create simple UI to control on/ff
 3. Create simple UI to control temperature

Bill of Material

No	Nama komponen	qty	Harga/pcs	link
1	Arus daya/Watt meter (PZEM 004T)	1	125-140K	Jual PZEM-004T SENSOR WATT VOLT CURRENT METER AC 100A 80-260V OPEN CT - Kota Bandung - CNC STORE BANDUNG Tokopedia
2	Ethernet shield (ENC28J60)	1	38k-60k	Jual ENC28J60 Ethernet Network Module - Modul LAN - Jakarta Utara - solarperfect Tokopedia
3	Sensor kelembaban	2	20-40k	Jual DHT22 Temperature and Humidity Module Modul Sensor Suhu dan Kelembaban - Jakarta Utara - solarperfect Tokopedia
4	AC to DC 12V 2A	1	40-100k	Jual AC to DC Converter Isolated 100-240V To 12V 2A Step Down Power Supply - 12V 2A - Kab. Purwakarta - Tekno Kreasi Elektronik Tokopedia
5	Arduino UNO	1	65-130k	Jual ARDUINO UNO R3 ATMEGA328P SMD CH340 COMPATIBLE BOARD PLUS USB CABLE - Kota Bandung - CNC STORE BANDUNG Tokopedia
6	IR LED Transmitter	4	500	Jual [CNC] INFRARED PHOTODIODE PHOTOTRANSISTOR EMITTER TRANSMITTER 940NM 5M - Kota Bandung - CNC STORE BANDUNG Tokopedia
7	TSOP1738	3	15-25k	Jual sensor remote tv besar tsop1738 - Jakarta Barat - PCM ELEKTRONIK JAKARTA Tokopedia
8	OLED	1	25-40k	https://tokopedia.link/VhoURWrR2Lb

Bill of Material

No	Nama komponen	qty	Harga/pcs	link
9	Step down	2	14k-20k	Jual XL4015 DC TO DC STEP DOWN BUCK CONVERTER 5A 0.8V-30V TO 5V-32V - Kota Bandung - CNC STORE BANDUNG Tokopedia
10	Project board	1		Jual [CNC] BREADBOARD MB-102 SOLDERLESS 830 830P - Kota Bandung - CNC STORE BANDUNG Tokopedia
11	Jumper	1	30k	Jual PAKET LENGKAP JUMPER 10CM FOR BREADBOARD - Kota Bandung - CNC STORE BANDUNG Tokopedia
12	IR Receiver	3	900	Jual [CNC] HX1838 38kHz UNIVERSAL IR INFRARED RECEIVER - Kota Bandung - CNC STORE BANDUNG Tokopedia
13	PCB	1	14k	Jual PCB DOT MATRIX THRU HOLE SINGLE LAYER 9X15CM 9*15CM LUBANG BO LONG PLAT - Kota Bandung - CNC STORE BANDUNG Tokopedia
14	Sensor arus zmt	1	22k	https://tokopedia.link/VY46fa3Q2Lb
15	Stok kontak female	2	15k	Jual STOP KONTAK OB STOP/OB STEKER STOP KONTAK 16A 220V - Kota Bandung - CNC STORE BANDUNG Tokopedia
16	Kabel AC	1	35k	Jual KABEL POWER AC KOMPUTER TIGA LUBANG 1.5 METER - Kota Bandung - CNC STORE BANDUNG Tokopedia
17	Jack DC	2	1k	Jual JACK DC 5.5 X 2.1 MALE SOCKET DC ADAPTOR JACK PLUG BLACK - Kota Bandung - CNC STORE BANDUNG Tokopedia

Bill of Material

No	Nama komponen	qty	Harga/pcs	link
18	Jumper Tunggal	1	30k	Jual KABEL JUMPER TUNGGAL FOR BREADBOARD 350PCS - Kota Bandung - CNC STORE BANDUNG Tokopedia
19	Header male	5	1k	Jual PIN HEADER MALE STRIP DOUBLE ROW 2X40 2.54MM BLACK HITAM - Kota Bandung - CNC STORE BANDUNG Tokopedia
20	Header Female	5	1.2k	Jual PIN HEADER FEMALE STRIP SINGLE ROW 1X40 2.54MM BLACK HITAM - Kota Bandung - CNC STORE BANDUNG Tokopedia
21	Fuse holder			https://tokopedia.link/egYGdoTN2Lb
22	Fuse 10A	1 set		https://tokopedia.link/eYmruCPN2Lb
23	Spacer 6mm	20	1k	https://tokopedia.link/m47hD6YN2Lb
24	Spacer 20mm	6	3k	https://tokopedia.link/ba5hk6FO2Lb
25	Rocker Swtich	1	3k	https://tokopedia.link/zsOFEIIO2Lb
26	Timah	1	31k	https://tokopedia.link/h6JxHssO2Lb
27	Pump Solder	1	20k	https://tokopedia.link/NjngZQuO2Lb
28	Solder Wick	1	15k	https://tokopedia.link/biywbdxO2Lb
29	BOX X5	1	12k	https://tokopedia.link/E5WghTKO2Lb

Bill of Material

No	Nama komponen	qty	Harga/pcs	link
30	ESP32	1	79K	https://tokopedia.link/1shSxZ1O2Lb
31	TTL sinyal converter	3	3K	https://tokopedia.link/bHZVYzuP2Lb
32	Resistor 330 ohm	2	1k	https://tokopedia.link/vGH1MfAP2Lb
33	Push on locking	2	6k	https://tokopedia.link/slGN1zPP2Lb
34	Push on momen pcb	1		https://tokopedia.link/iooeTvTP2Lb
35	Push on moment	2	3k	https://tokopedia.link/0EEy3jWP2Lb
36	Terminal blok cabut pasang(4p)	1	10k	https://tokopedia.link/xAcjMxLQ2Lb
37	Kabel micro usb	1	10k	https://tokopedia.link/zX2hyIER2Lb